Photoelectric Effect

Introduction

The photoelectric effect is the emission of electrons from the surface of a metal when electromagnetic radiation (such as visible or ultraviolet light) of the right frequency shines on the metal. At the time of its discovery, the classical wave model for light predicted that the energy of the emitted electrons would increase as the intensity (brightness) of the light increased.

Instead it was discovered that the energy of the emitted electrons was directly proportional to the frequency of the incident light, and that no electrons would be emitted if the light source was not above a certain threshold frequency. Lower energy electrons were emitted when light with relatively low frequency was incident on the metal, and higher energy electrons were emitted when light with relatively high frequency was incident on the metal.

The AP-8209 Photoelectric Effect Apparatus consists of a mercury light source enclosure, a photodiode tube enclosure, a base, the photoelectric effect test instrument, miscellaneous cords and cables, a power supply for the mercury light source and the test instrument, and a box of apertures, filters, caps, and alignment screws.

The apparatus has several important features:

- The current amplifier has high sensitivity and is very stable in order to improve the accuracy of the measurement.
- The photoelectric tube has low levels of dark current and anode reverse current.
- The optical filters are of high quality in order to avoid an error due to interference between different spectral lines.

Background Information

Many people contributed to the discovery and explanation of the photoelectric effect. In 1865 James Clerk Maxwell predicted the existence of electromagnetic waves and concluded that light itself was just such a wave. Experimentalists attempted to generate and detect electromagnetic radiation and the first clearly successful attempt was made in 1886 by Heinrich Hertz. In the midst of his experimentation, he discovered that the spark produced by an electromagnetic receiver was more vigorous if it was exposed to ultraviolet light. In 1888 Wilhelm Hallwachs demonstrated that a negatively charged gold leaf electroscope would discharge more rapidly than normal if a clean zinc disk connected to the electroscope was exposed to ultraviolet light. In 1899, J.J. Thomson determined that the ultraviolet light caused electrons to be emitted from the metal.

In 1902, Phillip Lenard, an assistant to Heinrich Hertz, used a high intensity carbon arc light to illuminate an emitter plate. Using a collector plate and a sensitive ammeter, he was able to measure the small current produced when the emitter plate was exposed to light. In order to measure the energy of the emitted electrons, Lenard charged the collector plate negatively so that the electrons from the emitter plate would be repelled. He found that there was a minimum "stopping" potential that kept all electrons from reaching the collector. He was surprised to discover that the "stopping" potential, V, - and therefore the energy of the emitted electrons - did *not* depend on the intensity of the light. He found that the maximum energy of the emitted electrons *did* depend on the color, or frequency, of the light.

In 1901 Max Planck published his theory of radiation. In it he stated that an oscillator, or any similar physical system, has a discrete set of possible energy values or levels; energies between these values never occur. Planck went on to state that the emission and absorption of radiation is associated with transitions or jumps between two energy levels. The energy lost or gained by the oscillator is emitted or absorbed as a quantum of radiant energy, the magnitude of which is expressed by the equation: $E = h\nu$ where E equals the radiant energy, ν is the frequency of the radiation, and h is a fundamental constant of nature. (The constant, h, became known as Planck's constant.)

In 1905 Albert Einstein gave a simple explanation of Lenard's discoveries using Planck's theory. The new 'quantum'-based model predicted that higher frequency light would produce higher energy emitted electrons (photoelectrons), independent of intensity, while increased intensity would only increase the number of electrons emitted (or photoelectric current). Einstein assumed that the light shining on the emitter material could be thought of as 'quanta' of energy (called photons) with the amount of energy equal to $h\nu$ with ν as the frequency. In the photoelectric effect, one 'quantum' of energy is absorbed by one electron. If the electron is below the surface of the emitter material, some of the absorbed energy is lost as the electron moves towards the surface. This is usually called the 'work function' (W_o). If the 'quantum' is more than the 'work function', then the electron is emitted with a certain amount of kinetic energy. Einstein applied Planck's theory and explained the photoelectric effect in terms of the quantum model using his famous equation for which he received the Nobel prize in 1921: $E = h\nu = KE_{max} + W_0$

Albert Einstein

where KE_{max} is the maximum kinetic energy of the emitted photoelectron. In terms of kinetic energy, $KE_{max} = h\nu - W_0$

If the collector plate is charged negatively to the 'stopping' potential so that electrons from the emitter don't reach the collector and the photocurrent is zero, the highest kinetic energy electrons will have energy eV where e is the charge on the electron and V is the 'stopping' potential.

$$eV = h\nu - W_0$$

$$V = \frac{h}{e}\nu - \frac{W_0}{e}$$

Einstein's theory predicts that if the frequency of the incident light is varied, and the 'stopping' potential, V, is plotted as a function of frequency, the slope of the line is h/e (see Figure 1).

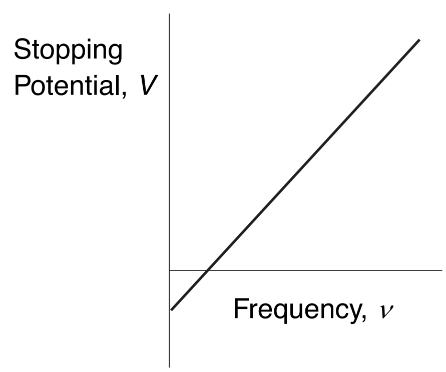
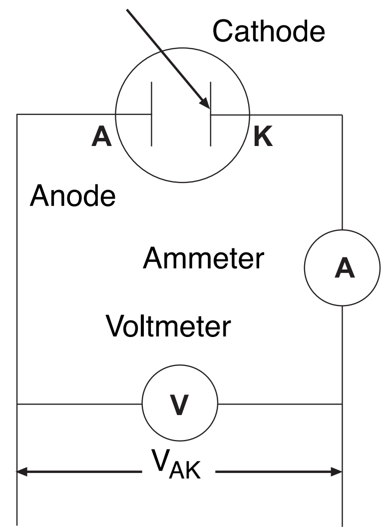


Figure 1: Stopping Potential, V versus Frequency, ν

Principle of the Experiment

When incident light shines on the cathode (K), photoelectrons can be emitted and transferred to the anode (A). This constitutes a photocurrent. By changing the voltage between the anode and cathode, and measuring the photocurrent, you can determine the characteristic current-voltage curves of the photoelectric tube.



The basic facts of the photoelectric effect experiments are as follows:

- For a given frequency (color) of light, if the voltage between the cathode and anode, V_{AK}, is equal to the stopping potential, V, the photocurrent is zero.
- When the voltage between the cathode and anode is greater than the stopping voltage, the photocurrent will increase quickly and eventually reach saturation. The saturated current is proportional to the intensity of the incident light. See Figure 2.
- Light of different frequencies (colors) have different stopping potentials. See Figure 3
- The slope of a plot of stopping potential versus frequency is the value of the ratio, h/e. See Figure 1.
- The photoelectric effect is almost instantaneous. Once the light shines on the cathode, photoelectrons will be emitted in less than a nanosecond.

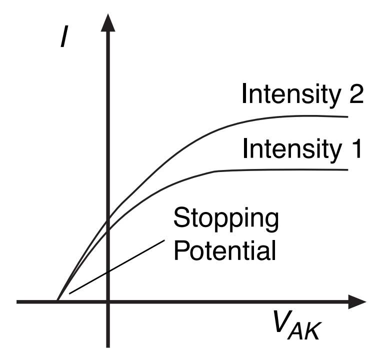


Figure 2: Current vs. Intensity

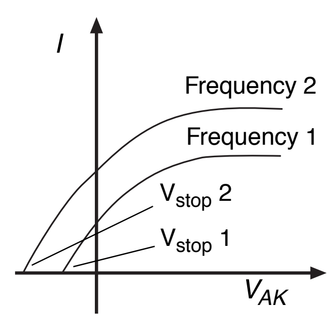


Figure 3: Current vs. Frequency

Experiment 1: Measuring and Calculating Planck's Constant, *h*

Preparation before measurement

1. Cover the window of the Mercury Light Source enclosure with the Mercury Lamp Cap from the Optical Filters box. Cover the window of the Photodiode enclosure with the Photodiode Cap from the Optical Filters box.
2. On the h/e Power Supply, turn on POWER and MERCURY LAMP. On the Photoelectric Effect Apparatus, push in the POWER button to the ON position.
3. Allow the light source and the apparatus to warm up for 20 minutes.

> ***Note:*** *It is very important to allow the light source and apparatus to warm up for 20 minutes prior to making any measurements.*

4. On the apparatus, set the VOLTAGE Range switch to –2 — 0 V. Turn the CURRENT RANGES switch to 10^{-13}.
5. To set the current amplifier to zero, first *disconnect* the 'A', 'K', and 'down arrow' (GROUND) cables from the back panel of the apparatus.
6. Press the PHOTOTUBE SIGNAL button in to CALIBRATION.
7. Adjust the CURRENT CALIBRATION knob until the current is zero.
8. Press the PHOTOTUBE SIGNAL button to MEASURE.
9. *Reconnect* the 'A', 'K', and 'down arrow' (GROUND) cables to the back of the apparatus.

Measurement

1. Uncover the window of the Photodiode enclosure. Place the 4 mm diameter aperture and the 365 nm filter onto the window of the enclosure. (See the sidebar note.)

> ***Note:*** *Always have a filter on the window of the Photodiode enclosure, and put the cap on the Mercury Light source whenever you change the filter or aperture. Never let the light from the Mercury Light source shine directly into the Photodiode enclosure.*

2. Uncover the window of the Mercury Light Source. Spectral lines of 365 nm wavelength will shine on the cathode in the phototube.
3. Adjust the VOLTAGE ADJUST knob until the current on the ammeter is zero.
4. Record the *magnitude* of the stopping potential for the 365 nm wavelength in Table 1.
5. Cover the window of the Mercury Light Source.

6. Replace the 365 nm filter with the 405 nm filter.

7. Uncover the window of the Mercury Light Source. Spectral lines of 405 nm wavelength will shine on the cathode in the phototube.

8. Adjust the VOLTAGE ADJUST knob until the current on the ammeter is zero.

9. Record the *magnitude* of the stopping potential for the 405 nm wavelength in Table 1.

10. Cover the window of the Mercury Light Source.

11. Repeat the measurement procedure for the other filters. Record the *magnitude* of the stopping potential for each wavelength in Table 1.

Table 1: Stopping Potential of Spectral Lines, 4 mm diameter Aperture

Item	1	2	3	4	5
Wavelength, λ (nm)	365.0	404.7	435.8	546.1	577.0
Frequency, $\nu = c/\lambda$, (x 10^{14} Hz)	8.214	7.408	6.879	5.490	5.196
Stopping Potential, V (V)					

Calculating

1. Plot a graph of Stopping Potential (V) versus Frequency (x 10^{14} Hz).

Note: For information on using the DataStudio program to plot the graph, see Appendix B.

2. Find the slope of the best-fit line through the data points on the Stopping Potential (V) versus Frequency (x 10^{14} Hz) graph.

Note: *DataStudio allows you to enter your data as ordered pairs in a Table display and then plot the data in a Graph display.*

Note: The slope is the ratio of *h/e*, so Planck's constant, *h,* is the product of the charge of the electron (e = 1.602 x 10^{-19} C) and the slope of the best-fit line.

According to the theory of linear regression, the slope of the Stopping Potential versus Frequency graph can be calculated using the following equation:

$$\textbf{slope} = \frac{\bar{\nu} \cdot \bar{\mathbf{V}} - \overline{(\nu \cdot \mathbf{V})}}{\bar{\nu}^2 - \overline{\nu^2}}$$

where $\bar{\nu} = \frac{1}{n}\sum_{i=1}^{n} \nu_i$, $\overline{\nu^2} = \frac{1}{n}\sum_{i=1}^{n} \nu_i^2$, $\bar{\mathbf{V}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{V}_i$, and $\overline{\nu \cdot \mathbf{V}} = \frac{1}{n}\sum_{i=1}^{n} \nu_i \cdot \mathbf{V}_i$.

3. Record the calculated slope and use it to calculate the value of Planck's constant, *h*.

Slope = ___________________

h = e x slope = ________________________

4. Estimate the error in the slope and round your result to the appropriate value. Compare your calculated value of h to the accepted value, h_0, 6.626 x 10^{-34} J s.

$$\textbf{percent difference} = \left|\frac{h - h_0}{h_0}\right| \times 100$$

5. Record your percent difference _______________________

Questions

1. How does your calculated value of h compare to the accepted value?
2. What do you think may account for the difference – if any – between your calculated value of h and the accepted value?
3. How can you find the value of the Work Function from the graph of Stopping Potential versus Frequency?

Extension

Repeat the data measurement and analysis procedure for the other two apertures in the OPTICAL FILTERS box

Table 2: .Stopping Potential of Spectral Lines, 2 mm diameter Aperture

Item	1	2	3	4	5
Wavelength, λ (nm)	365.0	404.7	435.8	546.1	577.0
Frequency, ν, (x 10^{14} Hz)	8.214	7.408	6.879	5.490	5.196
Stopping Potential, V (V)					

Table 3: .Stopping Potential of Spectral Lines, 8 mm diameter Aperture

Item	1	2	3	4	5
Wavelength, λ (nm)	365.0	404.7	435.8	546.1	577.0
Frequency, ν, (x 10^{14} Hz)	8.214	7.408	6.879	5.490	5.196
Stopping Potential, V (V)					

Questions

1. How does your calculated value of h for each different aperture compare to the accepted value, h_0, 6.626 x 10^{-34} J s?

2. How does light intensity affect the Stopping Potential?

Experiment 2: Measuring Current-Voltage Characteristics of Spectral Lines - Constant Frequency, Different Intensity

This section outlines the instructions for measuring and comparing the current versus voltage characteristics of one spectral line at three different light intensities.

Preparation for Measurement

1. Cover the window of the Mercury Light Source enclosure with the Mercury Lamp Cap from the Optical Filters box. Cover the window of the Photodiode enclosure with the Photodiode Cap from the Optical Filters box.

2. On the h/e Power Supply, turn on POWER and MERCURY LAMP. On the Photoelectric Effect Apparatus, push in the POWER button to the ON position.

3. Allow the light source and the apparatus to warm up for 20 minutes.

Note: *It is very important to allow the light source and apparatus to warm up for 20 minutes prior to making any measurements.*

4. On the apparatus, set the VOLTAGE Range Switch to –2 — +30 V. Turn the CURRENT RANGES Switch to 10^{-11}.

5. To set the current amplifier to zero, first disconnect the ‘A’, ‘K’, and ‘down arrow’ (GROUND) cables from the back panel of the apparatus.

6. Press the PHOTOTUBE SIGNAL button in to CALIBRATION.

7. Adjust the CURRENT CALIBRATION knob until the current is zero.

8. Press the PHOTOTUBE SIGNAL button to MEASURE.

9. Reconnect the ‘A’, ‘K’, and ‘down arrow’ (GROUND) cables to the back of the apparatus.

Measurement - Constant Frequency, Different Intensities

2 mm Aperture

1. Uncover the window of the Photodiode enclosure. Place the 2 mm diameter aperture and the 436 nm filter in the window of the enclosure.

2. Uncover the window of the Mercury Light Source enclosure. A spectral line of 436 nm will shine on the cathode in the Photodiode enclosure.

Note: *Always have a filter on the window of the Photodiode enclosure, and put the cap on the Mercury Light source whenever you change the filter or aperture. Never let the light from the Mercury Light source shine directly into the Photodiode enclosure.*

3. Adjust the –2—+30 V VOLTAGE ADJUST knob so that the current display is zero. Record the voltage and current in Table 4.

4. Increase the voltage by a small amount (for example, 1 V). Record the new voltage and current in Table 4.

5. Continue to increase the voltage by the same small increment. Record the new voltage and current each time in Table 4. Stop when you reach the end of the VOLTAGE range.

4 mm Aperture

1. Cover the window of the Mercury Light Source enclosure.

2. On the Photodiode enclosure, replace the 2 mm diameter aperture with the 4 mm diameter aperture. Put the 436 nm filter back onto the window.

3. Uncover the window of the Mercury Light Source enclosure. A spectral line of 436 nm will shine on the cathode in the Photodiode enclosure.

4. Adjust the –2—+30 V VOLTAGE ADJUST knob so that the current display is zero. Record the voltage and current in Table 4.

5. Increase the voltage by a small amount (e.g., 1 V) and record the new voltage and current in Table 4. Continue to increase the voltage by the same small increment and record the new voltage and current each time in Table 4. Stop when you reach the end of the VOLTAGE range

8 mm Aperture

1. Cover the window of the Mercury Light Source enclosure.
2. On the Photodiode enclosure, replace the 4 mm diameter aperture with the 8 mm diameter aperture. Put the 436 nm filter back onto the window.
3. Uncover the window of the Mercury Light Source enclosure. A spectral line of 436 nm will shine on the cathode in the Photodiode enclosure.
4. Adjust the –2—+30 V VOLTAGE ADJUST knob so that the current display is zero. Record the voltage and current in Table 4.
5. Increase the voltage by a small amount (e.g., 1 V) and record the new voltage and current in Table 4. Continue to increase the voltage by the same small increment and record the new voltage and current each time in Table 4. Stop when you reach the end of the VOLTAGE range.
6. Turn off the POWER on the apparatus.Turn off the MERCURY LAMP power switch and the POWER switch on the power supply. Return the apertures, filters, and caps to the OPTICAL FILTERS box.

Table 4: Current and Voltage of Spectral Lines

λ = 435.8 nm 2 mm dia.	V (V)												
	I (x 10^{-11} A)												
λ = 435.8 nm 4 mm dia.	V (V)												
	I (x 10^{-11} A)												
λ = 435.8 nm 8 mm dia.	V (V)												
	I (x 10^{-11} A)												

Analysis

1. Plot the graphs of Current (y-axis) versus Voltage (x-axis) for the one spectral line, 436 nm, at the three different intensities.

Questions

1. How do the curves of current versus voltage for the one spectral line at three different intensities compare? In other words, how are the curves similar to each other?
2. How do the curves of current versus voltage for the one spectral line at three different intensities contrast? In other words, how do the curves differ from each other.

Experiment 3: Measuring Current-Voltage Characteristics of Spectral Lines - Different Frequencies, Constant Intensity

This section outlines the instructions for measuring and comparing the current versus voltage characteristics of three spectral lines, 436 nm, 546 nm, and 577 nm, but with the same light intensity.

Preparation for Measurement

1. Cover the window of the Mercury Light Source enclosure with the Mercury Lamp Cap from the Optical Filters box. Cover the window of the Photodiode enclosure with the Photodiode Cap from the Optical Filters box.
2. On the h/e Power Supply, turn on POWER and MERCURY LAMP. On the Photoelectric Effect Apparatus, push in the POWER button to the ON position.
3. Allow the light source and the apparatus to warm up for 20 minutes.
4. On the apparatus, set the VOLTAGE Range Switch to –2 — +30 V. Turn the CURRENT RANGES Switch to 10^{-11}.
5. To set the current amplifier to zero, first disconnect the ‘A’, ‘K’, and ‘down arrow’ (GROUND) cables from the back panel of the apparatus.
6. Press the PHOTOTUBE SIGNAL button in to CALIBRATION.
7. Adjust the CURRENT CALIBRATION knob until the current is zero.
8. Press the PHOTOTUBE SIGNAL button to MEASURE.
9. Reconnect the ‘A’, ‘K’, and ‘down arrow’ (GROUND) cables to the back of the apparatus.

Measurement - Different Frequencies, Constant Intensity

436 nm Wavelength

1. Uncover the window of the Photodiode enclosure. Place the 4 mm diameter aperture and the 436 nm filter in the window of the enclosure.
2. Uncover the window of the Mercury Light Source enclosure. A spectral line of 436 nm will shine on the cathode in the Photodiode enclosure.
3. Adjust the –2—+30 V VOLTAGE ADJUST knob so that the current display is zero. Record the voltage and current in Table 4.

Note: *Always have a filter on the window of the Photodiode enclosure, and put the cap on the Mercury Light source whenever you change the filter or aperture. Never let the light from the Mercury Light source shine directly into the Photodiode enclosure.*

4. Increase the voltage by a small amount (for example, 1 V). Record the new voltage and current in Table 5.

5. Continue to increase the voltage by the same small increment. Record the new voltage and current each time in Table 5. Stop when you reach the end of the VOLTAGE range.

546 nm Wavelength

1. Cover the window of the Mercury Light Source enclosure.
2. On the Photodiode enclosure, replace the 436 nm filter with the 546 nm filter.

3. Uncover the window of the Mercury Light Source enclosure. A spectral line of 546 nm will shine on the cathode in the Photodiode enclosure.

4. Adjust the –2—+30 V VOLTAGE ADJUST know so that the current display is zero. Record the voltage and current in Table 5.

5. Increase the voltage by a small amount (e.g., 1 V) and record the new voltage and current in Table 5. Continue to increase the voltage by the same small increment and record the new voltage and current each time in Table 5. Stop when you reach the end of the VOLTAGE range

577 nm Wavelength

1. Cover the window of the Mercury Light Source enclosure.
2. On the Photodiode enclosure, replace the 546 nm filter with the 577 nm filter.

3. Uncover the window of the Mercury Light Source enclosure. A spectral line of 577 nm will shine on the cathode in the Photodiode enclosure.

4. Adjust the –2—+30 V VOLTAGE ADJUST know so that the current display is zero. Record the voltage and current in Table 5.

5. Increase the voltage by a small amount (e.g., 1 V) and record the new voltage and current in Table 54. Continue to increase the voltage by the same small increment and record the new voltage and current each time in Table 5. Stop when you reach the end of the VOLTAGE range.

6. Turn off the POWER on the apparatus.Turn off the MERCURY LAMP power switch and the POWER switch on the power supply. Return the apertures, filters, and caps to the OPTICAL FILTERS box.

Table 5: Current and Voltage of Spectral Lines

λ = 435.8 nm 4 mm dia.	V (V)												
	I (x 10^{-11} A)												
λ = 546.1 nm 4 mm dia.	V (V)												
	I (x 10^{-11} A)												
λ = 577.0 nm 4 mm dia.	V (V)												
	I (x 10^{-11} A)												

Analysis

1. Plot the graphs of Current (y-axis) versus Voltage (x-axis) for the three spectral lines, 436 nm, 546 nm, and 577 nm, at the one intensity.

Questions

1. How do the curves of current versus voltage for the three spectral lines at a constant intensity compare? In other words, how are the curves similar to each other?
2. How do the curves of current versus voltage for the three spectral lines at a constant intensity contrast? In other words, how do the curves differ from each other.